Mahmoud Elbatouty

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LinkedIn | GitHub | Portfolio

EDUCATION

Ph.D. in Computing, Intelligent Systems (Artificial Intelligence), University of North Florida — **Anticipated May 2028**
M.S. in Computing and Information Sciences, Data Science, University of North Florida | GPA 4.0/4.0 — **May 2025**
B.S. in Computing and Information Sciences, Data Science, University of North Florida | GPA 3.94/4.0 — **May 2023**
Minor in Statistics | *Summa Cum Laude* | *Hicks Honors College* | *Dean's List*

PROFESSIONAL EXPERIENCE

Data Scientist / Software Developer — Florida Blue — **May 2023 – Present**
Jacksonville, FL

- Built an unsupervised clustering pipeline across 50+ health-equity dimensions in Python and R; benchmarked K-Means, hierarchical, and density-based methods and applied PCA to produce 3 clusters still used in health-equity initiatives.
- Refactored 20+ Python and R codebases into reusable internal libraries for visualization, data transformation, and reporting, reducing repetitive analysis work across the team.
- Migrated 120+ jobs, processes, and reports to a consolidated Microsoft reporting server using Python, Git, and Jenkins, enabling legacy server decommissioning and lower operating costs.
- Automated financial reporting from Microsoft SQL Server stored procedures to secured Power BI dashboards; developed 50+ executive dashboards with React, JavaScript, Python, SQL, and Power BI.
- Migrated tables, stored procedures, and security controls from SQL Server to Microsoft Fabric and Snowflake using Spark, and supported on-premises generative AI pipelines.

AI Engineer — Overclock — **2026**
Remote

- Architected AI agents with defined goals, system instructions, memory, context management, tool permissions, triggers, guardrails, evaluation criteria, and human-in-the-loop escalation for reliable autonomous execution.
- Built Model Context Protocol (MCP) workflows connecting Claude and coding agents to APIs, databases, browser tools, and development systems; created reusable skills and structured prompts for tool selection, retrieval, and multi-step orchestration.
- Designed multi-agent and bot-based automations that decomposed operational workflows into specialized agents with shared context, task handoffs, failure recovery, observability, and approval checkpoints.

Modeling and Analytics Engineer — NLP Logix — **Sep 2022 – Dec 2022**
Jacksonville, FL

- Automated Census web scraping with Selenium and BeautifulSoup and built 14+ Python models with pandas and scikit-learn to forecast Florida child and maternal health outcomes, delivered through Tableau dashboards.

Data Scientist — Florida Data Science for Social Good — **Jun 2022 – Aug 2022**
Jacksonville, FL

- Mined 15M+ Florida voter records to classify statewide voting behavior from 2014–2020; built Tableau dashboards adopted across 29 League of Women Voters chapters and contributed to a peer-reviewed publication.

SELECTED PROJECT

Heirs' Property Risk Prediction — LISC AI4Good Hackathon, 3rd Place — **Sep 2024**

- Trained Random Forest, Logistic Regression, and neural network models to predict parcel-level tax-certificate outcomes (AUC-ROC 0.80); LISC Jacksonville used the results to allocate $70K and support 75+ families.

PUBLICATION

Elbatouty, M., et al. (2023). *Finding Insights in Florida Voter Participation*. Advances in Computing Research, LNNS 700, 44–55. DOI

TECHNICAL SKILLS

AI and Machine Learning: Python, R, SQL, scikit-learn, TensorFlow, PyTorch, Keras, Claude API, LLMs, RAG, AI agents, multi-agent systems, tool calling, MCP, prompt engineering, NLP, deep learning, time-series forecasting, statistical modeling, A/B testing

Data and Cloud: Supabase/Postgres, Microsoft SQL Server, Snowflake, Microsoft Fabric, Spark, Azure, AWS, SSIS, ETL/ELT, data modeling, stored procedures

Engineering and BI: Next.js, React, TypeScript, JavaScript, Vercel, Git, Jenkins, CI/CD, Docker, Power BI, Tableau, SSRS, Plotly, Jupyter